

```
#include <stdint.h>
```

```
#include <stdio.h>
```

```
void G(uint8_t v1, uint8_t v2, uint8_t v3, uint8_t v4){
```

```
    uint8_t a=v1;
```

```
    uint8_t b=v2;
```

```
    uint8_t c=v3;
```

```
    uint8_t d=v4;
```

```
    v1=d;
```

```
    v2=c;
```

```
    v3=a;
```

```
    v4=b;
```

```
}
```

```
    #include <stdio.h>
```

```
#include <stdint.h>
```

```
// AES S-box
```

```
uint8_t s[256] = {
```

```
    0x63,0x7c,0x77,0x7b,0xf2,0x6b,0x6f,0xc5,0x30,0x01,0x67,0x2b,0xfe,0xd7,0xab,0x76,  
    0xca,0x82,0xc9,0x7d,0xfa,0x59,0x47,0xf0,0xad,0xd4,0xa2,0xaf,0x9c,0xa4,0x72,0xc0,  
    0xb7,0xfd,0x93,0x26,0x36,0x3f,0xf7,0xcc,0x34,0xa5,0xe5,0xf1,0x71,0xd8,0x31,0x15,  
    0x04,0xc7,0x23,0xc3,0x18,0x96,0x05,0x9a,0x07,0x12,0x80,0xe2,0xeb,0x27,0xb2,0x75,  
    0x09,0x83,0x2c,0x1a,0x1b,0x6e,0x5a,0xa0,0x52,0x3b,0xd6,0xb3,0x29,0xe3,0x2f,0x84,  
    0x53,0xd1,0x00,0xed,0x20,0xfc,0xb1,0x5b,0x6a,0xcb,0xbe,0x39,0x4a,0x4c,0x58,0xcf,  
    0xd0,0xef,0xaa,0xfb,0x43,0x4d,0x33,0x85,0x45,0xf9,0x02,0x7f,0x50,0x3c,0x9f,0xa8,  
    0x51,0xa3,0x40,0x8f,0x92,0x9d,0x38,0xf5,0xbc,0xb6,0xda,0x21,0x10,0xff,0xf3,0xd2,  
    0xcd,0x0c,0x13,0xec,0x5f,0x97,0x44,0x17,0xc4,0xa7,0x7e,0x3d,0x64,0x5d,0x19,0x73,  
    0x60,0x81,0x4f,0xdc,0x22,0x2a,0x90,0x88,0x46,0xee,0xb8,0x14,0xde,0x5e,0x0b,0xdb,  
    0xe0,0x32,0x3a,0x0a,0x49,0x06,0x24,0x5c,0xc2,0xd3,0xac,0x62,0x91,0x95,0xe4,0x79,  
    0xe7,0xc8,0x37,0x6d,0x8d,0xd5,0x4e,0xa9,0x6c,0x56,0xf4,0xea,0x65,0x7a,0xae,0x08,  
    0xba,0x78,0x25,0x2e,0x1c,0xa6,0xb4,0xc6,0xe8,0xdd,0x74,0x1f,0x4b,0xbd,0x8b,0x8a,
```

```

0x70,0x3e,0xb5,0x66,0x48,0x03,0xf6,0x0e,0x61,0x35,0x57,0xb9,0x86,0xc1,0x1d,0x9e,
0xe1,0xf8,0x98,0x11,0x69,0xd9,0x8e,0x94,0x9b,0x1e,0x87,0xe9,0xce,0x55,0x28,0xdf,
0x8c,0xa1,0x89,0x0d,0xbf,0xe6,0x42,0x68,0x41,0x99,0x2d,0x0f,0xb0,0x54,0xbb,0x16
};

```

```

void H(uint8_t *v1, uint8_t *v2, uint8_t *v3, uint8_t *v4)

```

```

{
    uint8_t a = *v1;
    uint8_t b = *v2;
    uint8_t c = *v3;
    uint8_t d = *v4;

    a = s[(a << 3) & 0xFF]; // v1 = S(v1 << 3)
    b = s[a ^ b];           // v2 = S(v1 XOR v2)
    c = s[b ^ c];           // v3 = S(v2 XOR v3)
    d = s[c ^ d];           // v4 = S(v3 XOR v4)

    *v1 = a;
    *v2 = b;
    *v3 = c;
    *v4 = d;
}

```

```

int main()

```

```

{
    uint8_t v1 = 0x12;
    uint8_t v2 = 0x34;
    uint8_t v3 = 0x56;
    uint8_t v4 = 0x78;

    printf("Before: %02X %02X %02X %02X\n", v1, v2, v3, v4);
}

```

```
H(&v1, &v2, &v3, &v4);
```

```
printf("After : %02X %02X %02X %02X\n", v1, v2, v3, v4);
```

```
return 0;
```

```
}
```